

Express.js kya hai (short & clear)

👉 Express.js = Node.js ka **framework**

👉 Yeh tumhari life easy karta hai:

- Routing simple
- Middleware system
- Clean API structure

👉 Raw Node vs Express:

- Raw Node ❌ messy
- Express ✅ clean + scalable

1. Routing (API endpoints banana)

💡 Simple:

Routing ka matlab hai: “kis URL par kya kaam hoga”

 Example:

```
const express = require('express');  
const app = express();
```

```
app.get('/', (req, res) => {  
  res.send("Home API");  
});
```

```
app.get('/users', (req, res) => {  
  res.send("Users list");  
});
```

```
app.listen(3000);
```

Real use (Flutter app):

- /login
- /signup
- /users
- /donors

👉 Har route ek specific kaam karega

2. Middleware (Express ka MOST IMPORTANT concept)

Simple:

Middleware = request ke beech ka function

Flow:

Request → Middleware → Route → Response

Example:

```
app.use((req, res, next) => {  
  console.log("Request aayi hai");  
  next(); // next step  
});
```

Real use:

- Authentication check
- Logging
- Validation

Auth middleware example:

```
function auth(req, res, next) {  
  if (req.headers.token === "123") {  
    next();  
  } else {  
    res.send("Unauthorized");  
  }  
}
```

```
app.get('/profile', auth, (req, res) => {  
  res.send("User profile");  
});
```

✓ Route tab hi chalega jab auth pass ho

3. REST APIs (Industry Standard)

REST ka matlab:

Standard way APIs banane ka

Basic methods:

Method	Kaam
--------	------

GET	data lena
POST	data bhejna
PUT	update
DELETE	delete

Example:

```
app.get('/users', (req, res) => {  
  res.send("All users");  
});
```

```
app.post('/users', (req, res) => {  
  res.send("User created");  
});
```

Real backend (tumhara case):

- GET → donors list
- POST → register donor
- PUT → update profile
- DELETE → remove user

4. Request & Response Handling

Express me 2 main cheezen hoti hain:

- req → client se data
- res → client ko response

Example:

```
app.post('/login', (req, res) => {  
  const email = req.body.email;  
  res.send(`Welcome ${email}`);  
});
```

Important cheezen:

Request:

- req.body → POST data
- req.params → URL params
- req.query → query params

Response:

```
res.send("text");  
res.json({ name: "Arslan" });  
res.status(200).send("OK");
```

Real Flow (Flutter + Express)

Flutter → POST /login → Express → DB → Response → Flutter

Ek mini real example (combine sab kuch)

```
const express = require('express');  
const app = express();
```

```
app.use(express.json()); // middleware

// route
app.post('/login', (req, res) => {
  const { email, password } = req.body;

  if (email === "test@gmail.com" && password === "123") {
    res.json({ message: "Login success" });
  } else {
    res.status(401).json({ message: "Invalid" });
  }
});

app.listen(3000);
```

Tumhare liye Final Understanding

- ✓ Routing → API endpoints
- ✓ Middleware → beech ka control system
- ✓ REST → standard API design
- ✓ Req/Res → data ka flow

Reality Tip (Industry Level)

👉 Express sirf start hai
Real apps me tum use karoge:

- Controllers
- Services
- Database (MongoDB)
- Auth (JWT)